GENERATOR-BASED DELOOPING CONSTRUCTIONS FOR PRESENTED GROUPS IN HOMOTOPY TYPE THEORY

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complete omitted-proof appendix retained.

Allowed standard prerequisites (editorial).

- Homotopy type theory with univalence, higher inductive types, groupoid truncation and free-group universal properties. Author hott.tex and complete source-local constructions/proof appendix retained; standard results are cited to the HoTT Book. No independent Agda execution or correctness certification is performed.

2. Homotopy type theory

We unfortunately do not have enough space here to provide an introduction from scratch to dependent type theory and homotopy type theory, so we refer the reader to the reference book for an in depth presentation [27]. The main purpose of this section is to fix some terminology and notations for classical notions.

Universe. We write $\mathcal{U}$ for the universe, i.e. the large type of all small types, which we suppose to be closed under dependent sums and products. We write $\Pi(x : A).B$ or $(x : A) \to B$ for Π-types, and $A \to B$ for the case where B is non-dependent. Similarly, we write $\Sigma(x : A).B$ for Σ-types, and $A \times B$ for the non-dependent version. The two projections from a Σ-type are respectively written π and π'.

Paths. Given a type $A : \mathcal{U}$ and two elements $a, b : A$, we write $a =_A b$ for the type of identities, or paths, between a and b: its elements are proofs of equality between a and b. In particular, for any $a : A$, the type $a = a$ contains the term refl_a witnessing for reflexivity of equality. We sometimes write $x := t$ to indicate that x and t are equal by definition. The elimination principle of identities, aka path induction and often noted J, roughly states that, given $a : A$, in order to show a property $P : (x : A) \to (a = x) \to \mathcal{U}$ for every $x : A$ and $p : a = x$ it is enough to show it in the case where $x := a$ and $p := \mathrm{refl}_a$. By path induction, the following can be shown. Given a type A and a type family $B : A \to \mathcal{U}$, a path $p : x = y$ in A induces a function $B_p^{\to} : B(x) \to B(y)$ witnessing for the fact that equality is substitutive. As a special case, any path $p : A = B$ between two types $A, B : \mathcal{U}$ induces a function $p^{\to} : A \to B$, called the transport along p, as well as an inverse function $p^{\leftarrow} : B \to A$. Finally, given a function $f : A \to B$, any path $p : x = y$ in A induces a path $f^{=}(p) : f(x) = f(y)$ witnessing for the fact that equality is a congruence.

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Higher inductive types. Many functional programming languages allow the definition of inductive types, which are freely generated by constructors. For instance, the type S^0 of booleans is generated by two elements (true and false). In the context of homotopy type theory, languages such as cubical Agda feature a useful generalization of such types, called higher inductive types. They allow, in addition to traditional constructors for elements of the type, constructors for equalities between elements of the type. For instance, the type corresponding to the circle S^1 can be defined as generated by two points a and b and two equalities $p, q : a = b$ between those points. Higher-dimensional spheres S^n can be defined in a similar way.
Univalence. A map $f : A \to B$ is an equivalence when it admits both a left and a right inverse. In particular, every isomorphism is an equivalence. We write $A \simeq B$ for the type of equivalences from A to B. The identity is clearly an equivalence and we thus have, by path induction, a canonical map $(A = B) \to (A \simeq B)$ for every types A and B: the univalence axiom states that this map is itself an equivalence. In particular, every equivalence $A \simeq B$ induces a path $A = B$. It is known that univalence implies the function extensionality principle [27, Section 2.9]: given functions $f, g : A \to B$, if $f(x) = g(x)$ for any $x : A$ then $f = g$ (and the expected generalization to dependent function types is also valid).
Homotopy levels. A type A is contractible when the type $\Sigma(x : A).(y : A) \to (x = y)$ is inhabited: this means that we have a "contraction point" $a_0 : A$, and a continuous family of paths from a_0 to every other point in A. A type A is a proposition (resp. a set, resp. a groupoid) when $(x = y)$ is contractible (resp. a proposition, resp. a set) for every $x, y : A$. Intuitively, a contractible type is a point (up to homotopy), a proposition is a point or is empty, a set is a collection of points and a groupoid is a space which bears no non-trivial 2-dimensional (or higher) structure. We write Set for the type of sets. Given a type A, we write $\mathrm{isSet}(A)$ (resp. $\mathrm{isGroupoid}(A)$) for the predicate indicating that A is a set (resp. groupoid).
Truncation. Given a type A, its propositional truncation turns it into a proposition in a universal way. It consists of a type $\|A\|_{-1}$, which is a proposition, equipped with a map $|-|_{-1} : A \to \|A\|_{-1}$ such that, for any proposition B, the map $(\|A\|_{-1} \to B) \to (A \to B)$ induced by precomposition by $|-|_{-1}$ is an equivalence. Intuitively, the type $\|A\|_{-1}$ behaves like A, except that we do not have access to its individual elements: the elimination principle for propositional truncation states that in order to construct an element of B from an element of $\|A\|_{-1}$, we can only assume that we have an element of A if B itself is a proposition. The set truncation $\|A\|_0$ of a type A is defined similarly, as the universal way of turning A into a set, and we write $|x|_0$ for the image of $x : A$ in the truncation; and we can similarly define the groupoid truncation $\|A\|_1$.
Fibers. Given a function $f : A \to B$, we write $\mathrm{fib}_f\, b$ for the type $\Sigma(a : A).(f\, a = b)$, called the fiber of f at b. The function f is said to be surjective when the type $(b : B) \to \|\mathrm{fib}_f\, b\|_{-1}$ is inhabited, i.e. when every element of B merely admits a preimage.

3. Delooping groups

The external point of view. A group consists of a set A, together with an operation $m : A \to A \to A$ (the multiplication), an element $e : A$ (the unit), and an operation $i : A \to A$ (the inverse) such that multiplication

is associative, admits e as unit, and $i(x)$ is the two-sided inverse of any element $x : A$. We write Group for the type of all groups, and $G \to_{\mathrm{Grp}} H$ for the type of group morphisms between groups G and H. In the following, we use the traditional notations for groups: given two elements $x, y : G$, we simply write xy instead of $m(x, y)$, 1 instead of e, and x^{-1} instead of $i(x)$.

The internal point of view. A pointed type consists of a type A together with a distinguished element, often written $\star$ and sometimes left implicit. Given a pointed type $(A, \star)$, its loop space $\Omega\, A$ is defined as the type of paths from $\star$ to itself: $\Omega\, A := (\star = \star)$. The elements of this type are called loops. By path induction one can construct, for every two paths $p : a = b$ and $q : b = c$, a path in $a = c$ called their concatenation and written $p \cdot q$. Similarly, every path $p : a = b$, admits an inverse path $p^- : b = a$. When A is a pointed groupoid, $\Omega\, A$ is a set, and these operations canonically equip this set with a structure of group [27, Section 2.1].

Delooping groups. A delooping of a group G is a pointed connected groupoid $\mathrm{B}\, G$ together with an identification $\mathrm{d}_G : \Omega\, \mathrm{B}\, G = G$ (we recall that a type A is connected when the type $\|A\|_0$ is contractible, i.e. A has one connected component). The notation is justified by the fact that deloopings are unique. For instance, it is known that the circle is a delooping of $\mathbb{Z}$: indeed, S^1 is a connected groupoid, and its fundamental group is $\mathbb{Z}$ [27, Section 8.1].

4. Delooping using higher inductive types

Delooping as a higher inductive type. Given a group G, its delooping should have a point $\star$ and a loop for every element of the group. Moreover, we should ensure that the multiplication of G coincides with the concatenation operation on the loop space, and that the type we obtain is a (pointed connected) groupoid. This suggests considering a higher inductive type, noted $\mathrm{K}(G, 1)$, with the following constructors

$$\begin{array}{ll} \star & : \mathrm{K}(G,1) \\ \mathsf{loop} & : G \to \star = \star \\ \mathsf{loop\text{-}comp} & : (x, y : G) \to \mathsf{loop}\, x \cdot \mathsf{loop}\, y = \mathsf{loop}(xy) \\ \mathsf{trunc} & : \mathrm{isGroupoid}(\mathrm{K}(G,1)) \end{array}$$

This construction was first proposed by Finster and Licata. They also showed, using the encode-decode method, that it is a delooping of the original group, i.e. $\Omega\, \mathrm{K}(G, 1) = G$, see [17, Theorem 3.2]. Note that we only ask here that loop preserves multiplication (with loop-comp), because it can be shown that this implies preservation of unit and inverses. In particular, preservation of unit (see `EM.loop-id`) renders superfluous one of the constructors present in the original definition [17].

In the following, we will define a variant of this higher inductive type when the group G is presented, which is smaller and gives rise to computations closer to traditional group theory.

Presentations of groups. Given a set X, we write X^* for the free group over X [27, Theorem 6.11.6]. There is an inclusion function $\iota : X \to X^*$ which, by precomposition, induces an equivalence between morphisms of groups $X^* \to G$ and functions $X \to G$. We write $f^* : X^* \to G$ for the group morphism thus induced by a function $f : X \to G$. The elements of X^* can be described as formal composites $a_1 \dots a_n$ where each a_i is

an element of X or a formal inverse of an element of X (such that an element with an adjacent formal inverse cancel out).

Any free group X^* admits a delooping as a wedge of an X-indexed familly of circles. The corresponding type $\bigvee_X \mathrm{S}^1$ can be described as the coequalizer

$$X \rightrightarrows 1 \dashrightarrow \bigvee_X \mathrm{S}^1 \tag{1}$$

or, equivalently, as the higher inductive type generated by the two constructors $\star : \bigvee_X \mathrm{S}^1$ and $\mathsf{loop} : X \to \star = \star$.

Proposition 1. We have $\Omega \bigvee_X \mathrm{S}^1 = X^*$, i.e. the above type is a $\mathrm{B}\, X^*$.

Proof. The fact that $\bigvee_X \mathrm{S}^1$ is a delooping of X^* is not too difficult to show when X has decidable equality, see [27, Exercise 8.2] and [15], but the general case is more involved and was recently proved in [30]: the main issue is to show that this type is a groupoid. □

A group presentation $\langle X \mid R \rangle$ consists of a set X of generators, a set R of relations, and two functions $\pi, \pi' : R \to X^*$ respectively associating to a relation its source and target. We often write $r : u \Rightarrow v$ for a relation r with u as source and v as target. Given such a presentation P, the corresponding presented group $[P]$ is the set quotient X^*/R of the free group on X under the smallest congruence identifying the source and the target of every relation $r : R$. This type can be described as the type $[P] := \|X^* /\!\!/ R\|_0$ obtained by taking the set truncation of the coequalizer

$$R \underset{\pi'}{\overset{\pi}{\rightrightarrows}} X^* \overset{\kappa}{\dashrightarrow} X^* /\!\!/ R$$

From this also follows a description of $[P]$ as a higher inductive type:

$$\begin{aligned} \mathsf{word} &: X^* \to [P] \\ \mathsf{rel} &: (r : R) \to \mathsf{word}(\pi(r)) = \mathsf{word}(\pi'(r)) \\ \mathsf{trunc} &: \mathrm{isSet}([P]) \end{aligned}$$

A smaller delooping. Suppose given a group G along with a presentation $P := \langle X \mid R \rangle$, i.e. such that $G = [P]$. We define the type $\mathrm{B}\, P$ as the following higher inductive type:

$$\begin{aligned} \star &: \mathrm{B}\, P \\ \mathsf{gen} &: X \to (\star = \star) \\ \mathsf{rel} &: (r : R) \to (\mathsf{gen}^*(\pi(r)) = \mathsf{gen}^*(\pi'(r))) \\ \mathsf{trunc} &: \mathrm{isGroupoid}(\mathrm{B}\, P) \end{aligned}$$

This type is generated by a point $\star$, then the constructor gen adds a loop $\underline{a} : \star = \star$ for every generator a, the constructor rel adds an equality $\underline{a}_1 \cdot \underline{a}_2 \cdot \ldots \cdot \underline{a}_n = \underline{b}_1 \cdot \underline{b}_2 \cdot \ldots \cdot \underline{b}_m$ for each relation $a_1 \ldots a_n \Rightarrow b_1 \ldots b_m$, and the constructor trunc formally takes the groupoid truncation of the resulting type. Note that, because of the presence of $\mathsf{gen}^* : X^* \to (\star = \star)$ in the type of rel, the above inductive type is not accepted as is in standard proof assistants such as Agda. However, a definition can be done in two stages, by first considering $\bigvee_X \mathrm{S}^1$ (i.e. the type generated only by $\star$ and gen), and then defining a second inductive type further quotienting this type (i.e. adding the constructors rel and trunc), see `EM.Delooping`. Also, the definition of gen^* requires the group structure on $\bigvee_X \mathrm{S}^1$: the group

operations are easily defined from operations on paths (reflexivity, concatenation, symmetry), but the fact that it is a groupoid is non-trivial (see Theorem 1). Our main result in this section is the following:

Theorem 2 (`EM.theorem`)**.** Given a presentation $P := \langle X \mid R \rangle$, the type $\mathrm{B}\,P$ is a delooping of the group $[P]$.

Proof. By induction on $\mathrm{B}\,P$, we can define a function $f : \mathrm{B}\,P \to \mathrm{K}([P], 1)$ such that $f\star := \star$, and $f(\mathsf{gen}\,a) := \mathsf{loop}[a]$ for all $a : X$. It can be shown that f is then such that $f^{=}(\mathsf{gen}^*\,u) = \mathsf{loop}[u]$, for any $u : X^*$. We can therefore define the image $f^{=}(r)$ on a relation $r : u \Rightarrow v$ as the composite of equalities

$$f^{=}(\mathsf{gen}^*\,u) = \mathsf{loop}[u] = \mathsf{loop}[v] = f^{=}(\mathsf{gen}^*\,v)$$

where the equality in the middle follows from the fact that we have $[u] = [v]$ because of the relation r.

In the other direction, the group morphism $\mathsf{gen}^* : X^* \to \Omega\,\mathrm{B}\,P$ preserves relations (by rel), and thus induces a quotient morphism $g' : [P] \to \Omega\,\mathrm{B}\,P$. We can thus consider the function $g : \mathrm{K}([P], 1) \to \mathrm{B}\,P$ such that $g(\star) = \star$, for $x : [P]$ we have $g^{=}(\mathsf{loop}\,x) = g'(x)$, and for $x, y : [P]$ the image of $\mathsf{loop\text{-}comp}\,x\,y$ is canonically induced by the fact that g' preserves group multiplication.

Since $\mathrm{K}([P], 1)$ is a groupoid, in order to show that $f(g(x)) = x$ for every $x : \mathrm{K}([P], 1)$, it is enough to show that it holds for $x := \star$, which is the case by definition of f and g, and that this property is preserved under $\mathsf{loop}\,x$ for $x : [P]$, which follows from the fact that we have $f^{=}(g'(x)) = \mathsf{loop}\,x$ for any $x : [P]$ (this is easily shown by induction on x). Conversely, we have to show that $g(f(x)) = x$ holds for $x : \mathrm{B}\,P$. Again, this is shown by induction on x. □

As an interesting remark, the careful reader will note that the fact that the types X and R are sets does not play a role in the proof: in fact, those assumptions can be dropped here. Also, note that we do not need the choice of a representative in X^* for every element of $[P]$ in order to define the function g from gen^* in the above proof: intuitively, this is because the induced function g does not depend on such a choice of representatives. Finally, we should mention here that a similar result is mentioned as an exercise in [27, Example 8.7.17]; the proof suggested there is more involved since it is based on a generalized van Kampen theorem.

Example 3. The dihedral group D_5, see Theorem 13, admits the presentation

$$\langle r, s \mid r^4 = srs, sr^2s = r^3, rsr = s, r^3s = sr^2, sr^3 = r^2s, s^2 = 1 \rangle$$

Hence, by Theorem 2 we can construct a delooping of D_5 as an higher inductive type generated by two loops (corresponding to r and s) and six 2-dimensional cells (corresponding to the relations). Note that this is much smaller than $\mathrm{K}(D_5, 1)$ (it has 2 instead of 10 generating loops, and 6 instead of 100 relations), thus resulting in shorter proofs when reasoning by induction.

Example 4. Any group G admits a presentation, the standard presentation, with one generator $\underline{a}$ for every element $a : G$, and relations $\underline{a}\;\underline{b} = \underline{ab}$ for every pair of generators, as well as $\underline{1} = 1$. By applying Theorem 2, we actually recover the inductive type $\mathrm{K}(G, 1)$ as delooping of G.

5. Delooping with torsors

In this section, we recall the other classical approach to constructing deloopings of groups by using G-torsors, which originates in classical constructions of algebraic topology [12]. Most of the material of the section is already known, for which reason proofs are not much detailed. A more in-depth presentation can be found in recent works such as [2, 5].

Group actions. Given a group G and a set A, an action of G on A is a group morphism from G to $A \simeq A$, that is a map $\alpha : G \to (A \simeq A)$ such that

$$\alpha(xy) = \alpha(x) \circ \alpha(y) \qquad \alpha(1) = \mathrm{id}_A \tag{2}$$

for all $x, y : G$.

A G-set is a set equipped with an action of G, and we write Set_G for the type of G-sets. We often simply denote a G-set by the associated action α and write $\mathrm{dom}(\alpha)$ for the set on which G acts.

Lemma 5 (`GSetProperties.isGroupoidGSet`)**.** The type Set_G is a groupoid.

Proof. The type of sets is a groupoid [27, Theorem 7.1.11]. Given a set A, the type of functions $A \to A$ is a set [27, Theorem 7.1.9] and thus a groupoid. Finally, the axioms Eq. (2) of actions are propositions (because A is a set) and thus groupoids. We conclude since groupoids are closed under Σ-types [27, Theorem 7.1.8]. □

Given G-sets α and β, a morphism between them consists of a function $f : \mathrm{dom}\,\alpha \to \mathrm{dom}\,\beta$ which preserves the group action, in the sense that for every $x : G$ and $a : \mathrm{dom}\,\alpha$, we have

$$\beta(x)(f(a)) = f(\alpha(x)(a)). \tag{3}$$

A morphism which is also an equivalence is called an isomorphism and we write $\alpha \simeq^{\mathrm{Set}_G} \beta$ for the type of isomorphisms between α and β. We write $\mathrm{Aut}(\alpha)$ for the type of automorphisms $\alpha \simeq^{\mathrm{Set}_G} \alpha$, which is a group under composition. The equalities between G-sets can be conveniently characterized as follows.

Proposition 6 (`GSetProperties.GSet≡Decomp`)**.** Given two G-sets α and β, an equality between them consists of an equality $p : \mathrm{dom}\,\alpha = \mathrm{dom}\,\beta$ such that the function induced by transport along p, namely $p^{\to} : \mathrm{dom}\,\alpha \to \mathrm{dom}\,\beta$, is a morphism of G-sets.

Proof. The characterization of equalities between Σ-types [27, Theorem 2.7.2] entails that an equality between $(\mathrm{dom}\,\alpha, \alpha)$ and $(\mathrm{dom}\,\beta, \beta)$ is a pair consisting of an equality $p : \mathrm{dom}\,\alpha = \mathrm{dom}\,\beta$ and an equality $q : p^{\to}(\alpha) = \beta$ (we can forget about the equality between the components expressing the properties required for group actions since those are propositions). By [27, Lemma 2.9.6] and function extensionality, we finally have that the type of q is equivalent to the type $\beta(x) \circ p^{\to} = p^{\to} \circ \alpha(x)$. □

It easily follows from this proposition that any equality between G-sets induces an isomorphism of G-sets, as customary for equalities between algebraic structures [27, Section 2.14]. In fact, this map from equalities to isomorphisms can itself be shown to be an equivalence:

Proposition 7 (`GSetProperties.GSetPath`)**.** Given G-sets α and β, the canonical function

$$(\alpha = \beta) \to (\alpha \simeq^{\mathrm{Set}_G} \beta)$$

is an equivalence. Moreover, given a G-set α, the induced equivalence

$$(\alpha = \alpha) \simeq (\alpha \simeq^{\mathrm{Set}_G} \alpha)$$

is compatible with the canonical group structures on both types.

Proof. This is actually an instance of a more general correspondence between equalities and isomorphisms of algebraic structures, which is known under the name of structure identity principle, see [11] and [27, Section 9.8], and can be understood as a generalisation of univalence for types having an algebraic structure. □

Torsors. For any group G, there is a canonical G-set called the principal G-torsor and noted P_G, corresponding to the action of G on itself by left multiplication. Moreover, its group of automorphisms is precisely the group G:

Proposition 8 (`Deloopings.PGloops`)**.** Given a group G, we have an equality of groups

$$(\mathrm{P}_G \simeq^{\mathrm{Set}_G} \mathrm{P}_G) = G$$

Proof. The two functions $\phi : \mathrm{Aut}\,\mathrm{P}_G \to G$ and $\psi : G \to \mathrm{Aut}\,\mathrm{P}_G$, respectively defined by $\phi(f) := f(1)$ and $\psi(x)(y) := yx$ are mutually inverse group morphisms, see Section A. □

The type Set_G is thus "almost" a delooping of G. Namely, it is a groupoid (Theorem 5), which is pointed by P_G and satisfies $\Omega\,\mathrm{Set}_G = G$ by Theorems 7 and 8. It only lacks being connected, which can easily be addressed. Given a pointed type A, its connected component $\mathrm{Comp}\,A$ is the type $\Sigma(x : A).\|\star = x\|_{-1}$. It is well-known that this type is pointed by $(\star, |\,\mathrm{refl}\,|_{-1})$, connected and has the same loop space as the original type, i.e. we have $\Omega\,\mathrm{Comp}\,A = \Omega\,A$, see Section C. We thus have:

Theorem 9 (`Deloopings.torsorDeloops`)**.** The connected component of P_G in Set_G, i.e. the type $\mathrm{Comp}(\mathrm{Set}_G, \mathrm{P}_G)$, is a delooping of G.

The elements of the connected component of the principal G-set are usually called G-torsors.

6. Generated torsors

Fix a group G. Given a set X and a map $\gamma : X \to G$, we say that X generates G (with respect to γ) when $\gamma^* : X^* \to G$ is surjective. From now on, we suppose that we are in such a situation. We now provide a variant for the construction of a delooping of G by G-torsors described in the previous section, taking advantage of the additional data of a generating set in order to obtain smaller and simpler constructions. Note that here, contrarily to Section 4, we only need a set of generators, not a full presentation.

Actions of sets. Given a type A, we write $\operatorname{End} A$ for its type of endomorphisms, i.e. maps $A \to A$. An action of the set X on a set A is a morphism $X \to \operatorname{End} A$, i.e. a family of endomorphisms of A indexed by X. We write Set_X for the type

$$\operatorname{Set}_X \quad := \quad \Sigma(A : \operatorname{Set}).(X \to \operatorname{End} A)$$

of actions of X. An element α of this type consists in a set $\operatorname{dom}\alpha$ with a function $\alpha : X \to \operatorname{End}(\operatorname{dom}\alpha)$ and is called an X-set. A morphism between X-sets α and β is a function $f : \operatorname{dom}\alpha \to \operatorname{dom}\beta$ satisfying Eq. (3) for every $x : X$. The identities between X-sets can be characterized in a similar way as for G-sets, see Theorem 6, and Theorem 7 also extends in the expected way.

Precomposition by γ induces a function $U : \operatorname{Set}_G \to \operatorname{Set}_X$ which can be thought of as a forgetful functor from G-sets to X-sets. Note that U depends on γ but we leave it implicit for concision.

Applications of the generated delooping. We have seen in the previous section that the connected component of the principal G-torsor P_G in G-sets is a delooping of G. Our aim in this section is to show here that this construction can be simplified by taking the connected component of the restriction of P_G to X-sets.

Before proving this theorem, which is formally stated as Theorem 11 below, we shall first illustrate its use on a concrete example. Consider $\mathbb{Z}_n$, the cyclic group with n elements. We write $s : \mathbb{Z}_n \to \mathbb{Z}_n$ for the successor (modulo n) function, which is an isomorphism. By Theorem 9, we know that the type

$$\Sigma(A : \operatorname{Set}_{\mathbb{Z}_n}).\| \operatorname{P}_{\mathbb{Z}_n} = A\|_{-1}$$

of $\mathbb{Z}_n$-torsors is a model of $\operatorname{B}\mathbb{Z}_n$. This type is the connected component of the principal $\mathbb{Z}_n$-torsor $\operatorname{P}_{\mathbb{Z}_n}$ in the universe $\operatorname{Set}_{\mathbb{Z}_n}$ of sets with an action of $\mathbb{Z}_n$, i.e. sets A equipped with a morphism $\alpha : \mathbb{Z}_n \to \operatorname{Aut} A$. Such a set A is thus comes with one automorphism $\alpha(k)$ for every element $k : \mathbb{Z}_n$, therefore k automorphisms in this case. However, most of them are superfluous: 1 generates all the elements of $\mathbb{Z}_n$ by addition, so $\alpha(1)$ generates all the $\alpha(k)$ by composition because $\alpha(k) = \alpha(1)^k$. The useful data of a $\mathbb{Z}_n$-set thus boils down to a set A together with one automorphism $\alpha : \operatorname{Aut} A$ such that $\alpha^n = \operatorname{id}_A$.

Indeed, writing $\operatorname{Set}^{\circlearrowleft} := \Sigma(A : \operatorname{Set}).\operatorname{End} A$ for the type of all endomorphisms (on any set), our theorem will imply that the type

$$\Sigma((A, f) : \operatorname{Set}^{\circlearrowleft}).\|(\mathbb{Z}_n, s) = (A, f)\|_{-1} \tag{4}$$

(the connected component of the successor modulo n in the universe of set endomorphisms) is still a delooping of $\mathbb{Z}_n$. Note that we didn't assume that f is an isomorphism nor that it should verify $f^n = \operatorname{id}$. This is because both properties follow from the fact that f is in the connected component of the successor (which satisfies those properties). Similarly, we do not need to explicitly assume that the domain of the endomorphism is a set.

Our theorem thus allows to define, in a relatively simple way, types corresponding to deloopings of groups. As recalled in the introduction, this is particularly useful when one is not disposed to use higher inductive types (e.g. because their definition, implementation and semantics are not entirely mature). This is in fact the reason why this approach was used in UnitMath to define the circle [3], and we provide a generic

way to similarly define other types. We expect that it can be used to reason about groups and compute invariants such as their cohomology [8, 6, 4]. On a side note, one might be worried by the fact that we are "biased" (by using a particular set of generators), which allows us to be more concise but might make more difficult generic proofs compared to G-torsors: we expect that this is not the case because in order to define the group G itself, one usually needs to resort to a presentation, and thus is also biased in some sense...
The generated delooping. In the following, we write P_X for $U\,\mathrm{P}_G$.

Proposition 10 (`XSetProperties.theorem`)**.** We have a group equivalence $\Omega\,\mathrm{P}_G \simeq \Omega\,\mathrm{P}_X$.

Proof. From Theorem 6, an element of $\Omega\,P_G$ consists of an equality $p : G = G$ in $\mathcal{U}$ such that

$$\mathrm{P}_G(x) \circ p^{\rightarrow} = p^{\rightarrow} \circ \mathrm{P}_G(x)$$

for every $x : G$. By function extensionality and the definition of the action P_G, this is equivalent to requiring, for every $g, z : G$ that

$$g(p^{\rightarrow}(z)) = p^{\rightarrow}(gz) \tag{5}$$

Note that the above equality is between elements of G, which is a set, and is thus a proposition. Similarly, an element of $\Omega\,\mathrm{P}_X$ consists of an equality $p : G = G$ in $\mathcal{U}$ satisfying

$$\gamma(x)(p^{\rightarrow}(z)) = p^{\rightarrow}(\gamma(x)z) \tag{6}$$

for every $x : X$ and $z : G$.

Clearly, any equality $p : G = G$ in $\Omega\,\mathrm{P}_G$ also belongs to $\Omega\,\mathrm{P}_X$ since the condition Eq. (6) is a particular case of Eq. (5). We thus have a function $\phi : \Omega\,\mathrm{P}_G \to \Omega\,\mathrm{P}_X$. Conversely, consider an element $p : G = G$ of $\Omega\,\mathrm{P}_X$, i.e. satisfying Eq. (6) for every $x : X$ and $z : G$. Our aim is to show that it belongs to $\Omega\,\mathrm{P}_G$. Given $g, z : G$, we thus want to show that Eq. (5) holds. Since γ^* is surjective, because X generates G, we know that there merely exists an element u of X^* such that $\gamma^*(u) = x$. Since Eq. (5) is a proposition, by the elimination principle of propositional truncation, we can actually suppose given such a u, and we have

$$\begin{aligned}
x(p^{\rightarrow}(y)) &= \gamma^*(u)(p^{\rightarrow}(y)) && \text{since } \gamma^*(u) = x \\
&= p^{\rightarrow}(\gamma^*(u)y) && \text{by repeated application of Eq. (6)} \\
&= p^{\rightarrow}(xy) && \text{since } \gamma^*(u) = x.
\end{aligned}$$

The second equality essentially corresponds to the commutation of the following diagram, where $u := x_1 x_2 \dots x_n$ with $x_i : X$:

$$\begin{array}{ccccccccccc}
G & \xrightarrow{\gamma(x_1)} & G & \xrightarrow{\gamma(x_2)} & G & \longrightarrow & \dots & \longrightarrow & G & \xrightarrow{\gamma(x_n)} & G \\
{\scriptstyle p^{\rightarrow}}\downarrow & & {\scriptstyle p^{\rightarrow}}\downarrow & & {\scriptstyle p^{\rightarrow}}\downarrow & & & & {\scriptstyle p^{\rightarrow}}\downarrow & & {\scriptstyle p^{\rightarrow}}\downarrow \\
G & \xrightarrow[\gamma(x_1)]{} & G & \xrightarrow[\gamma(x_2)]{} & G & \longrightarrow & \dots & \longrightarrow & G & \xrightarrow[\gamma(x_n)]{} & G
\end{array}$$

This thus induces a function $\psi : \Omega\,\mathrm{P}_X \to \Omega\,\mathrm{P}_G$. The functions ϕ and ψ clearly preserve the group structure (given by concatenation of paths) and are mutually inverse of each other, hence we have the equivalence we wanted. □

Theorem 11. The type $\mathrm{Comp}\,\mathrm{P}_X$ is a delooping of G.

Proof. Since taking the connected component preserves loops spaces (Theorem 21), we have that $\Omega\,\mathrm{Comp}\,\mathrm{P}_X$ is equal to $\Omega\,\mathrm{P}_X$, which in turn is equal to $\Omega\,\mathrm{P}_G$ by Theorem 10, and thus to G by Theorem 9. □

The delooping of G constructed in previous theorem is the component of P_X in X-sets:

$$\Sigma(A : \mathcal{U}).\Sigma(S : \mathrm{isSet}\,A).\Sigma(f : X \to \mathrm{End}\,A).\|\,\mathrm{P}_X = (A, S, f)\|_{-1}$$

Since for any type A, the type $\mathrm{isSet}\,A$ is a proposition [27, Theorem 7.1.10], and the underlying type of P_X is a set, the underlying type of any X-set in the connected component of P_X will also be a set. As a consequence, the above type can slightly be simplified, by dropping the requirement that A should be a set:

Proposition 12. The type $\Sigma(A : \mathcal{U}).\Sigma(f : X \to \mathrm{End}\,A).\|\,\mathrm{P}_X = (A, f)\|_{-1}$ is a delooping of G.

For instance, the delooping Eq. (4) of $\mathbb{Z}_n$ can slightly be simplified as

$$\Sigma((A, f) : \mathcal{U}^{\circlearrowleft}).\|(\mathbb{Z}_n, s) = (A, f)\|_{-1}$$

where $\mathcal{U}^{\circlearrowleft} := \Sigma(A : \mathcal{U}).\,\mathrm{End}\,A$ is the type of all endomorphisms of the universe.

Example 13. Theorem 11 applies to every group for which a generating set is known (and, of course, the smaller the better). For instance, given a natural number n, the dihedral group D_n is the group of symmetries of a regular polygon with n sides. It has $2n$ elements and is generated by two elements s (axial symmetry) and r (rotation by an angle of $2\pi/n$). Hence the connected component of the symmetry and the rotation in the type of pairs of set endomorphisms, i.e.

$$\Sigma(A : \mathrm{Set}).\Sigma(f, g : \mathrm{End}\,A \times \mathrm{End}\,A).\|(D_n, s, r) = (A, f, g)\|_{-1}$$

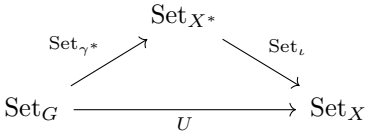
is a delooping of the dihedral group D_n.

Alternative proof. We would like to provide another proof Theorem 10, which was suggested by an anonymous reviewer. It is based on the idea that in order to show $\Omega\,\mathrm{P}_X \simeq \Omega\,\mathrm{P}_G$, it is enough to show that $U : \mathrm{Set}_G \to \mathrm{Set}_X$ is an embedding, i.e. that for every $\alpha, \beta : \mathrm{Set}_G$ the induced function $U^{=} : (\alpha = \beta) \to (U\alpha = U\beta)$ between path spaces is an equivalence [27, Definition 4.6.1]. It relies on the following result whose proof can be found in Section A.

Lemma 14. Given a type $A : \mathcal{U}$, type families $P, Q : A \to \mathcal{U}$ and $f : (a : A) \to P\,a \to Q\,a$, the map $\Sigma A.f : \Sigma A.P \to \Sigma A.Q$ is an embedding if and only if $f\,a : P\,a \to Q\,a$ is an embedding for every $a : A$.

Proposition 15. The map $U : \mathrm{Set}_G \to \mathrm{Set}_X$ is an embedding.

Proof. Given a morphism of groups $f : H \to K$, we write $\mathrm{Set}_f : \mathrm{Set}_K \to \mathrm{Set}_H$ for the function induced by precomposition. In particular, by definition, we have $U := \mathrm{Set}_\gamma$. The function $\gamma : X \to G$ can be decomposed as $\gamma^* \circ \iota$, and therefore Set_γ can be decomposed as $\mathrm{Set}_\iota \circ \mathrm{Set}_{\gamma^*}$:



Both maps are embeddings so that U is an embedding by composition. Namely,

- Set_{γ^*} is an embedding. Consider the map
$$F : (A : \mathrm{Set}) \to (G \to_{\mathrm{Grp}} \mathrm{Aut}\, A) \to (X^* \to_{\mathrm{Grp}} \mathrm{Aut}\, A)$$
obtained by precomposition by γ^*. Since γ^* is surjective we have that $F\, A$ is an embedding for every set A [27, Lemma 10.1.4]. By Theorem 14, we deduce that $\Sigma\, \mathrm{Set}\, .F$, which is Set_{γ^*}, is an embedding.
- Set_{ι} is an embedding. By universal property of X^*, given a set A, the map
$$(X^* \to_{\mathrm{Grp}} \mathrm{Aut}\, A) \to (X \to \mathrm{Aut}\, A)$$
obtained by precomposition by ι is an equivalence, and thus an embedding. Moreover, the property of being an isomorphism is a proposition, hence the forgetful map
$$(X \to \mathrm{Aut}\, A) \to (X \to \mathrm{End}\, A)$$
is an embedding because its fibers are propositions. Since embeddings are stable under composition, we deduce that the induced map
$$F : (A : \mathrm{Set}) \to (X^* \to_{\mathrm{Grp}} \mathrm{Aut}\, A) \to (X \to \mathrm{End}\, A)$$
is such that $F\, A$ is an embedding for every set A. By Theorem 14, the map $\Sigma\, \mathrm{Set}\, .F$, which is Set_{ι}, is thus an embedding. □

Appendix A. Omitted proofs

Proof of Theorem 8. The two functions
$$\begin{aligned} \phi : \mathrm{Aut}\, \mathrm{P}_G &\to G & \qquad \psi : G &\to \mathrm{Aut}\, \mathrm{P}_G \\ f &\mapsto f(1) & \qquad x &\mapsto y \mapsto yx \end{aligned}$$
are group morphisms. Namely, given $f, g : \mathrm{Aut}\, \mathrm{P}_G$, we have
$$\phi(g \circ f) = g \circ f(1) = g(f(1)1) = f(1)g(1) = \phi(f)\phi(g) \qquad \phi(\mathrm{id}) = \mathrm{id}(1) = 1$$
and given $x, y : G$, we have for every $z : G$,
$$\psi(xy)(z) = z(xy) = (zx)y = \psi(y) \circ \psi(x)(z) \qquad \psi(1)(x) = x1 = \mathrm{id}(x)$$
Moreover, they are mutually inverse. Namely, given $f : \mathrm{Aut}\, \mathrm{P}_G$ and $x : G$, we have
$$\psi \circ \phi(f)(x) = xf(1) = f(x1) = f(x) \qquad \phi \circ \psi(x) = 1x = x$$
We thus have $\mathrm{Aut}\, \mathrm{P}_G \simeq G$ and we conclude by univalence. □

Proof of Theorem 14. By definition, the map $\Sigma A.f$ is an embedding iff for every (a, x) and (a', x') in ΣAX, the induced map
$$(a, x) = (a', x') \to (a, f\, a\, x) = (a', f\, a'\, x')$$
is an equivalence. By the characterization of equalities in Σ-types [27, Theorem 2.7.2], this map corresponds to a map
$$(\Sigma(p : a = a').P_p^{\to}(x) = x') \to (\Sigma(p : a = a').Q_p^{\to}(f\, a\, x) = f\, a'\, x')$$
By [27, Theorem 4.7.7], this is an equivalence if and only if the fiber map
$$(P_p^{\to}(x) = x') \to (Q_p^{\to}(f\, a\, x) = f\, a'\, x')$$

is an equivalence for every $p : a = a'$. By path induction, this is true if and only if

$$(f\,a)^{=} : x = x' \to f\,a\,x = f\,a\,x'$$

is an equivalence for all $a : A$, and $x, x' : Xa$. By definition, this is the requirement that $f\,a$ is an embedding for all $a : A$. □

Appendix B. Equivalence between internal and external groups

Functoriality of delooping. One of the main properties of the delooping operation is that it is a "local inverse" to taking loop spaces in the following sense:

Proposition 16. Given pointed connected groupoids A and B and a group morphism $f : \Omega\,A \to \Omega\,B$, there is a unique pointed morphism $g : A \to B$ such that $\Omega\,g = f$.

Proof. By [29, Corollary 12], with $n = 0$, we have that the type $\Sigma(g : A \to B).\,\Omega\,g = f$ is equivalent to $(p, q : \star_A = \star_A) \to f(p \cdot q) = f(p) \cdot f(q)$. This type is a proposition because $\Omega\,B$ is a set (because B is a groupoid), and inhabited (because f is a morphism of groups), and thus contractible. □

As an immediate consequence of the above lemma, deloopings are unique:

Proposition 17. Given two deloopings $\mathrm{B}\,G$ and $\mathrm{B}'\,G$ of a group G, we have $\mathrm{B}\,G = \mathrm{B}'\,G$.

Given a group morphism $f : G \to H$ such that both G and H admit deloopings (and this actually always holds by Theorem 9), the delooping of f is the morphism

$$\mathrm{B}\,f : \mathrm{B}\,G \to \mathrm{B}\,H$$

associated, by Theorem 16, to the morphism $\mathrm{d}_H^{\leftarrow} \circ f \circ \mathrm{d}_G^{\rightarrow} : \Omega\,\mathrm{B}\,G \to \Omega\,\mathrm{B}\,H$. By Theorem 16, this operation is functorial in the sense that that it preserves identities and composition.

Equivalence between the two points of view. Although this is not central in this article, we shall mention here the fundamental equivalence provided by the above constructions; details can be found in [2]. We write $\mathrm{IntGroup}$ for the type of internal groups, i.e. pointed connected groupoids.

Theorem 18. The maps $\Omega : \mathrm{IntGroup} \to \mathrm{Group}$ and $\mathrm{B} : \mathrm{Group} \to \mathrm{IntGroup}$ form an equivalence of types.

Proof. Given a group G, we have $\Omega\,\mathrm{B}\,G = G$ by definition of $\mathrm{B}\,G$. Given an internal group A, we have $\mathrm{B}\,\Omega\,A \simeq A$ by Theorem 17. □

The above theorem thus states looping and delooping operators allow us to go back and forth between the external and the internal point of view of group theory in homotopy type theory. Note that the torsor construction only gives a delooping in a larger universe than the original group unless one makes additional assumptions such as the replacement axiom [25, Axiom 18.1.8].

Internal group actions. In a similar way that the traditional notion of group admits an internal reformulation (Section 3), the notion of action also admits an internal counterpart which can be defined as follows. Given a group G, an internal action of G on a set A is a function

$$\alpha : \mathrm{B}\, G \to \mathrm{Set}$$

such that $\alpha(\star) = A$. Since Set is a groupoid [27, Theorem 7.1.11], by Theorem 18, we have equivalences of types

$$(\mathrm{B}\, G \to \mathrm{Set}) \quad \simeq \quad (\Omega\, \mathrm{B}\, G \to \Omega(\mathrm{Set}, A)) \quad \simeq \quad (G \to \mathrm{Aut}\, A)$$

which show that internal group actions correspond to external ones: the delooping operator internalizes an external group action, and the looping operator externalizes an internal group action.

Appendix C. Connected components

We define the connected component of the pointed type A as the type of points which are merely connected to the distinguished point of A. This type is noted $\mathrm{Comp}\, A$ (or $\mathrm{Comp}(A, \star)$ when we want to specify the distinguished element $\star$). Formally,

$$\mathrm{Comp}\, A \quad := \quad \Sigma(x : A).\|\star = x\|_{-1}$$

This type is canonically pointed by $(\star, |\,\mathrm{refl}\,|_{-1})$. This construction deserves its name because it produces a connected space, whose geometry is the same as the original space around the distinguished point, as shown in the following two lemmas.

Lemma 19 (`Comp.isConnectedComp`)**.** The type $\mathrm{Comp}\, A$ is connected.

Proof. It can be shown that a type X is connected precisely when both $\|X\|_{-1}$ and $(x, y : X) \to \|x = y\|_{-1}$ are inhabited, i.e. when X merely has a point and any two points are merely equal [27, Exercise 7.6]. In our case, the type $\mathrm{Comp}\, A$ is pointed and thus $\|\,\mathrm{Comp}\, A\|_{-1}$ holds. Moreover, suppose that there are two points (x, p) and (y, q) in $\mathrm{Comp}\, A$ with $x, y : A$, $p : \|\star = x\|_{-1}$ and $q : \|\star = y\|_{-1}$. Our goal is to show that $\|(x, p) = (y, q)\|_{-1}$ holds, which is a proposition, so by elimination of propositional truncation, we can therefore assume that p (resp. q) has type $\star = x$ (resp. $\star = y$). Hence, we can construct a path $p^{-} \cdot q$ of type $x = y$, and therefore $(x, p) = (y, q)$ because the second components belong to a proposition by propositional truncation. We conclude that $\|(x, p) = (y, q)\|_{-1}$ and finally that $\mathrm{Comp}\, A$ is connected. □

Lemma 20 (`Comp.loopCompIsLoop`)**.** We have $\Omega\, \mathrm{Comp}\, A = \Omega\, A$.

Proof. We begin by showing that the type

$$\Sigma((x, t) : \mathrm{Comp}\, A).(\star = x) \tag{7}$$

is contractible. In order to do so, observe that we have the following equivalence of types:

$$\begin{aligned}\Sigma((x, t) : \mathrm{Comp}\, A).(\star = x) &\simeq \Sigma((x, t) : \Sigma(x : A).\|\star = x\|_{-1}).(\star = x) \\ &\simeq \Sigma(x : A).(\|\star = x\|_{-1}) \times (\star = x) \\ &\simeq \Sigma((x, p) : \Sigma(x : A).(\star = x)).\|\star = x\|_{-1}\end{aligned}$$

using classical associativity and commutativity properties of Σ-types. Moreover, the type $\Sigma(x : A).(\star = x)$ is contractible [27, Lemma 3.11.8], therefore the whole type on the last line is a proposition (as a sum of

propositions over a proposition), and therefore also the original type Eq. (7). We write $\star'$ for the element $(\star, \| \operatorname{refl}_\star \|_{-1})$ of $\operatorname{Comp} A$. The type Eq. (7) is pointed by the canonical element $(\star', \operatorname{refl})$ and thus contractible as a pointed proposition.

We have a morphism

$$\begin{array}{ccccc} F : ((x,t) : \operatorname{Comp} A) & \to & (\star' = (x,t)) & \to & (\star = x) \\ (x,t) & & p & \mapsto & \pi^=(p) \end{array}$$

sending a path p to the path obtained by applying the first projection. It canonically induces a morphism

$$\begin{array}{c} \Sigma((x,t) : \operatorname{Comp} A).(\star' = (x,t)) \to \Sigma((x,t) : \operatorname{Comp} A).(\star = x) \\ ((x,t), p) \mapsto ((x,t), \pi^=(p)) \end{array}$$

between the corresponding total spaces. Since the left member is contractible (by [27, Lemma 3.11.8] again) and the right member is contractible (as shown above), this is an equivalence. By [27, Theorem 4.7.7], for every $x : \operatorname{Comp} A$, the fiber morphism Fx is also an equivalence. In particular, with x being $\star'$, we obtain $\Omega \operatorname{Comp} A \simeq \Omega A$ (as a type) and we can conclude by univalence. Note that the equivalence preserves the group structure so that the equality also holds in groups. □

As a direct corollary of the two above lemmas, we have:

Proposition 21. Given a pointed groupoid A, $\operatorname{Comp} A$ is a delooping of ΩA.

Remark 22. Some people write $\operatorname{Aut} A$ for ΩA and the above proposition states that we have $\operatorname{B} \operatorname{Aut} A = \operatorname{Comp} A$. For this reason, the (confusing) notation $\operatorname{BAut} A$ is also found in the literature for $\operatorname{Comp} A$.

Appendix D. The flattening lemma

We recall here the classical flattening lemma, see [27, Section 6.12] for a more detailed presentation and proof.

Lemma 23 (Flattening for coequalizers)**.** Suppose given a coequalizer

$$A \underset{g}{\overset{f}{\rightrightarrows}} B \xrightarrow{h} C$$

with $p : h \circ f = h \circ g$, and a type family $P : C \to \mathcal{U}$. Then the diagram

$$\Sigma A.(P \circ h \circ f) \underset{\Sigma g.e}{\overset{\Sigma f.(\lambda_.\,\mathrm{id})}{\rightrightarrows}} \Sigma B.(P \circ h) \xrightarrow{\Sigma h.(\lambda_.\,\mathrm{id})} \Sigma C.P$$

is a coequalizer, where the map

$$e : (a : A) \to P(h(f(a))) \to P(h(g(a)))$$

is induced by transport along p by

$$e\, a\, x := P^{\to}_{p(a)}(x).$$

Note that there is a slight asymmetry: we could have formulated a similar statement with $\Sigma A.(P \circ h \circ g)$ as left object.

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